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The Five Equations

THE MATH THAT QUIETLY RUNS YOUR MONEY



The Five Equations That Quietly Run Your Entire Financial Life (And Why School Never Showed You Any)



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Reply

You were taught arithmetic in school and told it was math. Then you were taught "personal finance," if you were lucky, as a list of tips: save more, spend less, invest early. Nobody told you that underneath every one of those tips sits an equation. Nobody told you that these equations are short, free, decided almost every rich person's outcome before they were born, and take about an hour to actually understand.

Here is the uncomfortable truth. Money is not really a subject. It is applied mathematics wearing a suit. Every decision you will ever make about a salary, a mortgage, an investment, a fee, a loan, or a pension is being decided by five equations, whether you know them or not. The people who know them are not smarter than you. They are just holding a small piece of paper that you have been quietly kept away from your whole life.

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I want to hand you that piece of paper. Five equations. Each one is short enough to memorize on the walk to lunch. Each one explains a piece of the financial world that looks mysterious from the outside and becomes obvious the moment you can read the math underneath it.

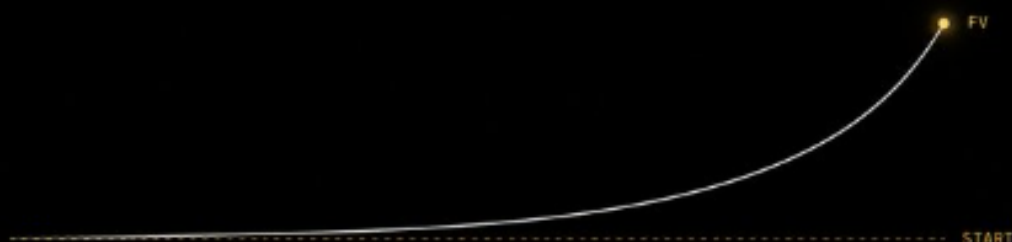
By the end, you will have the same core toolkit that a hedge fund analyst, a pension actuary, and a private banker all quietly rely on. Not their instincts, not their contacts, just their math. Which turns out to be your math too.

THE FIVE EQUATIONS AT A GLANCE

1. **Compound growth.** What a dollar becomes over time.
 2. **Rule of 72.** How fast money doubles at any rate.
 3. **Present value.** What a future dollar is worth right now.
 4. **Geometric mean.** Why the "average return" you were told is a lie.
 5. **Real return.** What your money actually did after inflation ate its share.
- Five formulas. Zero jargon. Together they explain roughly 90% of every honest financial decision anyone has ever made.*

Time bends money.

COMPOUND GROWTH - FIBONACCI, 1202



Equation 1: Compound Growth, or Why Time Beats Talent

The first and most important equation in the world of money looks like this:

$$FV = PV \times (1 + r)^n$$

That is it. Five symbols. Let me translate them into human.

- **FV** = future value. What your money will be worth later.
- **PV** = present value. What you have now.
- **r** = rate of return per period, written as a decimal (10% is 0.10).
- **n** = number of periods (usually years).

The whole thing reads: "your money in the future equals your money now, multiplied by one plus your rate, raised to the number of years."

The magic of this equation is in that little exponent. The n is what makes the whole world of finance strange. Without it, money would just grow in a straight line. With it, money grows in a curve that starts as a whisper and ends as a scream.

Concrete example. Put \$10,000 at 8% a year, add nothing more, and let it sit:

- Year 10: \$21,589
- Year 20: \$46,610
- Year 30: \$100,627
- Year 40: \$217,245
- Year 50: \$469,016

- Year 50: \$469,016

Notice the shape. Each decade doubles roughly. The first decade adds around . The last decade alone adds around . Same rate. Same original bet. The exponent is doing the work.

This is why a 25-year-old who invests \$10K and never adds another cent ends up with more money at 65 than a 40-year-old who invests \$10K every single year and gets to the same finish line. Not slightly more. Almost twice as much. The 25-year-old is not smarter or richer. They are just holding a bigger exponent.

Talent grows linearly. Time compounds. The math will always eventually favor time.

Equation 2: The Rule of 72, or the Only Mental Math Trick a Banker Actually Uses

Nobody wants to plug numbers into $FV = PV \times (1 + r)^n$ in their head at a dinner party. So bankers, actuaries, and quants use a shortcut that has been around since at least 1494, when the Italian mathematician Luca Pacioli wrote it down in the same book that introduced double-entry bookkeeping to Europe.

Doubling time (in years) $\approx 72 \div$ rate of return (in percent)

That is the whole trick. How long does money double at 6%? $72 \div 6 = 12$ years. At 9%? $72 \div 9 = 8$ years. At 3%? $72 \div 3 = 24$ years.

It is not perfectly precise, but it is accurate to within a percent or two for any normal rate, and it lets you do in three seconds what would otherwise take a spreadsheet.

Once you have this in your head, whole conversations change. Someone tells you a bond pays 4% a year. You immediately know: this money doubles every 18 years. Someone tells you inflation is running at 6%. You

Once you have this in your head, whole conversations change. Someone tells you a bond pays 4% a year. You immediately know: this money doubles every 18 years. Someone tells you inflation is running at 6%. You immediately know: prices double every 12 years, so what costs \$100 today costs \$200 in a decade. Someone offers you a "safe" 2% savings account. You immediately know: your money doubles every 36 years, which for most humans is once in an adult lifetime.

This is why professionals seem to have supernatural intuition about numbers. They do not. They just carry two mental shortcuts, and this is one of them.

Here is the same rule inverted, which almost nobody teaches. If something loses value at rate r , the rule tells you how fast it halves.

Inflation at 3%: money loses half its purchasing power in 24 years.

Inflation at 6%: half is gone in 12. Inflation at 9%: half is gone in 8.

The rule works in both directions. Rising and falling, growing and rotting. And it fits in your head.

72 solves it all.

THE DOUBLING RULE - PACIOLEI, 1494



THE FIVE EQUATIONS

Equation 3: Present Value, or What a Future Dollar Is Actually Worth

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Here is a trick question. Someone offers you \$1,000, either today or one year from now. Which is worth more?

Almost everyone answers "today," and almost everyone gives the wrong reason. They say something like "because I could spend it now." The real reason is much more useful, and it is the reason every acquisition, insurance policy, pension, mortgage, and lottery jackpot in the world is priced the way it is.

The reason is that \$1,000 today can be invested. If you have a 6% return available, then \$1,000 today becomes \$1,060 in a year. So \$1,000 today is not equal to \$1,000 next year. **It is equal to about \$1,060 next year, or equivalently, \$1,000 next year is only worth about \$943 today.**

This is called **present value**, and the equation is just the compound growth equation solved backwards.

$$PV = FV \div (1 + r)^n$$

Same five symbols. Same idea. You are asking: "what would I need to invest today at rate r to end up with this future amount in n years?"

The consequences of this one equation are enormous, and they explain a huge amount of the financial world.

Lottery jackpots. When a lottery announces a "\$300 million jackpot," they almost never have \$300 million in cash. They have enough to buy an annuity that will pay out \$300 million over 30 years. In present value terms, that jackpot is worth roughly half of what the billboard says. The lump-sum option is not the "reduced" prize. It is the honest one.

Insurance. When an insurance company quotes you a life policy, they are running present value on future payouts. If a policy will pay \$500,000 in 30 years and their assumed return is 5%, that future payout has a present value of roughly \$115,000. That is what they are actually reserving. Everything else is spread.

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Mortgages. When a bank quotes you a 30-year mortgage, they are calculating the present value of your future stream of payments. Understanding this is why the same 1% difference in interest rate can mean \$80,000 more or less paid over the life of the loan. The bank is not being greedy. They are pricing time.

Job offers. A \$150,000 salary starting today is worth more than a \$170,000 salary starting in three years, at any reasonable growth assumption. Present value is why "wait a few years for a raise" is usually a losing proposition, and why career switches early in life pay off more than they seem to.

A future dollar shrinks.

PRESENT VALUE - FISHER, 1907



THE FIVE EQUATIONS

The person who knows this equation looks at every future promise, from a bond coupon to a startup equity offer, and asks: what is this actually worth today? Everyone else takes the future number at face value, and everyone else quietly overpays.

Equation 4: The Geometric Mean, or Why Every "Average Return" You've Ever Been Sold Is Slightly a Lie

This is the equation the finance industry does not particularly want you to know. It is short. It is honest. It will change what you think of every fund advertisement you have ever seen.

Suppose a fund gives you these returns over three years: +50%, -50%, +50%.

Question. What is your average return?

The instinctive answer is: $(50 - 50 + 50) \div 3 = 16.7\%$. That looks great. A 16.7% average return sounds like a Warren Buffett-tier product.

The real answer is closer to +7%. And here is where it gets ugly.

Your actual money did this: \$100 → \$150 → \$75 → \$112.50. Over three years, you turned \$100 into \$112.50, which is a total gain of 12.5%, or roughly 4% per year compounded. Not 16.7%.

The number the industry loves to advertise is the **arithmetic mean**, the simple average. The number that describes what actually happened to your dollars is the **geometric mean**.

$$\text{Geometric mean} = ((1 + r_1) \times (1 + r_2) \times \dots \times (1 + r_n))^{(1/n)} - 1$$

You multiply the growth factors, take the nth root, and subtract 1. It sounds complicated, and it is one line in any spreadsheet:
`=GEOMEAN(...)-1.`

Here is the merciless rule underneath it:

The geometric mean is always less than or equal to the

The geometric mean is always less than or equal to the arithmetic mean. They are equal only when every year has the exact same return.

The more volatile the returns, the bigger the gap. A fund with returns of +30% and -30% has an arithmetic mean of 0% and a geometric mean of about -4.6%. On average, you did nothing. In reality, you lost money. Volatility itself has a cost, and that cost has a name: **variance drag**.

This is why a "smooth" 8% return is worth much more than a "wild" 8% return, even though the ad-brochure number is the same. It is why hedge funds that swing hard often disappoint their investors even when their pitch decks look brilliant. And it is why any honest performance report shows the **CAGR** (compound annual growth rate), which is just the geometric mean by another name.

The average is a lie.

GEOMETRIC MEAN - CAUCHY, 1821

ARITHMETIC 16.7%

GEOMETRIC 4%

THE FIVE EQUATIONS

Once you know this equation, you look at every fund's average return with a small extra question in your head: "arithmetic or geometric?" If they will not tell you, the answer is almost always the one they do not want you to compare.

Equation 5: Real Return, or the Invisible Line That Actually Runs Your Life

There is one final equation, and it is the one most people never think about, which is exactly why it eats them.

You put money in a savings account paying 4%. Inflation is 3%. What did your money actually do?

The intuitive answer is "grew by 4%." The correct answer is that it grew by roughly 1% in purchasing power. Your dollars multiplied, but each dollar now buys less. Your money grew nominally and shrank in real terms almost simultaneously.

$$\text{Real return} = ((1 + \text{nominal return}) \div (1 + \text{inflation})) - 1$$

You divide 1 plus your return by 1 plus inflation, and subtract 1. Or, for a quick approximation that works fine at small numbers:

$$\text{Real return} \approx \text{nominal return} - \text{inflation}$$

That is the equation. But the mindset it forces is the actual gift.

- A "guaranteed 5% bond" during 4% inflation is a 1% return, not 5%.
- A "0% checking account" during 3% inflation is a **-3% return**. You are being paid to lose money slowly.
- A stock market averaging 10% nominal during 3% inflation is a 7% real return. That 7% is the number that actually matters.

The reason this equation is so quietly important is that inflation is the one line item in your financial life that you never see on any statement. Your account balance goes up. Your credit card is paid. Everything looks fine. Meanwhile a slow, invisible tax is running in the background, taking two, three, five percent of your purchasing power every year.

The rich do not necessarily earn more nominal return than everyone else. They just make sure their nominal return is comfortably above inflation, always. Everyone else quietly gets poorer while their bank balance goes up.

Money you did not invest is not "safe." It is being taxed by inflation at a rate you never consented to.

Learning to think in real terms instead of nominal terms is the single mental switch that separates people who understand money from people who only handle it.

A future dollar shrinks.

PRESENT VALUE - FISHER, 1907



THE FIVE EQUATIONS

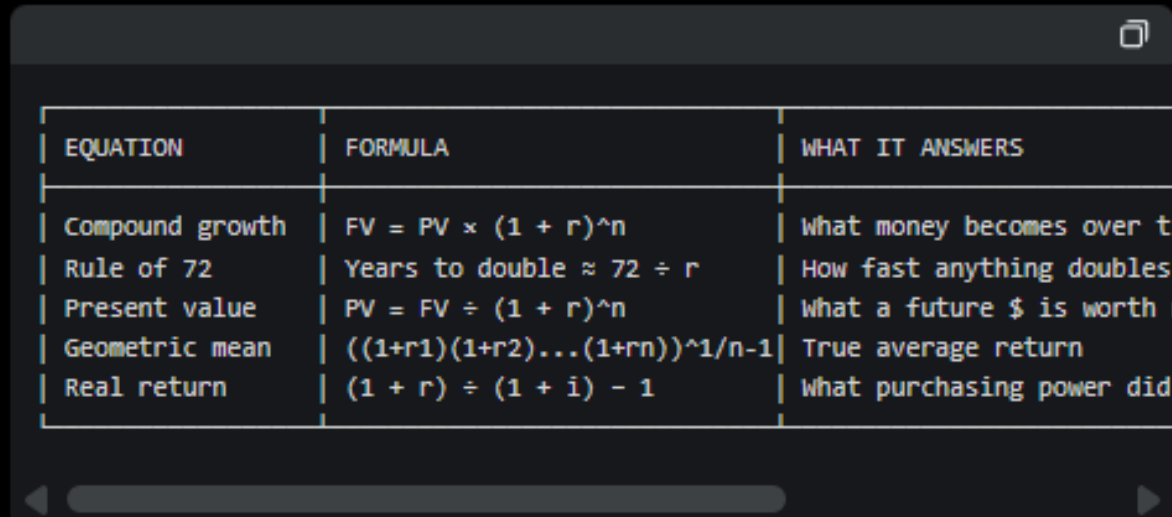
Putting the Five Equations Together

Each equation on its own is a tool. Together, they are a lens, and they let you see through almost any financial claim you will ever encounter.

Here is the entire toolkit in one table, with what each one actually does for you in real life.

Five equations. Two of them are just algebra rearranged. One is a mental shortcut from 1494. All five fit on one napkin.

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EQUATION	FORMULA	WHAT IT ANSWERS
Compound growth	$FV = PV \times (1 + r)^n$	What money becomes over t
Rule of 72	Years to double $\approx 72 \div r$	How fast anything doubles
Present value	$PV = FV \div (1 + r)^n$	What a future \$ is worth i
Geometric mean	$((1+r1)(1+r2)\dots(1+rn))^{1/n-1}$	True average return
Real return	$(1 + r) \div (1 + i) - 1$	What purchasing power did

And they let you do things almost no one around you can do:

- Estimate any doubling time in three seconds
- Read a fund's performance sheet without being fooled
- Price a future promise honestly
- Instantly separate nominal from real returns
- Understand why time is the most valuable asset you own

Why This Actually Matters

The financial industry does not want you to be bad at math. It wants you to be **selectively bad at math**. Good enough to sign contracts and pay fees. Not good enough to notice what the contracts are actually doing.

A 2% management fee sounds like a rounding error. Run it through the compound growth equation for 40 years and it eats close to half of your final wealth. You did not notice because 2% does not feel like half.

A "5% guaranteed" savings product during 4% inflation sounds safe. Run it through the real return equation and you earned 1%. You did not notice because 5% sounded like a lot.

A "20% average return hedge fund" sounds spectacular. Ask for the geometric mean instead of the arithmetic mean and you often find a much duller number. You did not notice because they only advertised the flattering one.

This is not a conspiracy. It is a feature. Every industry that sells complexity relies on its customers not being able to do the underlying math. Insurance, mortgages, funds, pensions, credit cards, all of them work best when the customer trusts the salesperson's summary instead of running the equation themselves.

The five equations above are the entire defence. Not against fraud, which is rare, but against the much more common problem of quietly bad deals dressed up in reassuring language.

What Actually Matters Here

Put it all together in one picture.

Money is not a mysterious force. It is a small set of equations that you were never quite handed. The people who look wealthy or wise about finance are not psychically gifted. They are running the same five formulas you just read, over and over, on every decision that crosses their desk. That is the entire "secret."

- Compound growth explains why time is more valuable than talent.
- The Rule of 72 explains why professionals can estimate anything in seconds.
- Present value explains why every future promise is worth less than it looks.
- Geometric mean explains why most advertised returns are quietly overstated.
- Real return explains why nominal safety is often real loss.

None of these formulas are hidden. None of them require a degree. None

None of these formulas are hidden. None of them require a degree. None of them cost anything. All of them are older than your grandparents. They were written down centuries ago by mathematicians and clerks who never imagined that their work would still be running the wealth of the world 500 years later.

The single most useful mental habit you can build is this: whenever you face a financial claim, before you feel anything about it, ask which of the five equations governs it, and run the number.

The rich are not people who beat the equations. They are people who never argued with them.

Here is the question worth sitting with.

If five short equations, all free, all public, all older than any bank on Earth, decide most of your financial life, and you have been navigating money without them for years, then the honest question is not "why don't I understand finance?" It is:

"How much has it already cost me to not know these five things, and how much more am I willing to let it cost me?"

The math is on the napkin. The rest is up to you.

If this piece just changed how you look at finance, follow – [@lumenxbt](#)

Cash quietly rots.

REAL RETURN – FISHER EQUATION, 1938

NOMINAL +5%

REAL +1%

↑
INFLATION +4%



Not because I need the numbers. Because what you just read is one topic out of roughly twenty that quietly rewire how you understand risk, wealth, and how decisions actually get made.

Every next piece here will take one topic and break it all the way down:

- Why 90% of active fund managers lose to a boring index (and what the math says about "expertise")
- The single formula that decides how much to bet on anything, and why nobody uses it
- Why insurance is priced by mathematicians, not marketers, and what that tells you about what to buy
- Why a 1% edge, applied correctly, is worth more than a 50% edge applied wrong

Every article works exactly like this one. No filler. Real math under simple words. An ending that leaves you seeing the world slightly differently than you did before.

If you have been looking for a place where difficult things get explained once, and correctly, you just found it. You will not stumble on a feed like this twice by accident. Unfollowing takes half a second. Nothing to lose, and a completely different way of seeing money to gain.

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